

When Solar Farms Become a Nuisance

With a Focus on the 'Renewable Energy 3020 Implementation Plan' in the Honam Region of South Korea

AMS MADE
Metropolitan Innovators

Yunsu Jang

WUR – 1724126, TU Delft – 6419909
yunsu.jang@wur.nl

21, 12, 2025

Abstract – Rapid solar deployment is widely framed as an unquestionably 'good', yet utility-scale expansion can generate substantial social and ecological side effects. This essay examines solar farm development in Korea's Honam region under the 2017-2025 '3020 Renewable Energy Implementation Project', asking what must be considered in planning and construction. Drawing on Korean news sources, policy studies, and theory, it analyses unintended outcomes through three integrated perspectives - spatial justice (who is excluded from inputs), a socio-technical (how 'function' is simplified), and (eco)system (flow of the overall system). International analysis from Zonnepark in the Netherlands highlight emerging mitigation efforts, while underscoring the need for slower, more inclusive, system-aware solar governance.

Keywords: Renewable Energy 3020 Implementation Project, South Korea, Honam region, Spatial Justice, Socio-technical, (Eco)System

Content

1 Introduction.....	3
2. Renewable Energy 3020 Implementation Project	4
3. Mechanism of Spatial Justice, Socio-technical and (Eco)system	5
4. Unexpected outputs of the 3020 Project – view from Spatial Justice, Socio-technical and (Eco)system.	7
5. Comparative Insights, Limitations, and Directions for Further Research	9
5.1. Beyond Honam, Zonnepark in the Netherlands.....	9
5.2. Limitation & Further research recommendation	10
6. Conclusion.....	10
7. References	11

1 Introduction

Countries are racing to install solar panels. Global PV capacity increased from 402.5GW in 2017, to more than 2.2TW at the beginning of 2025 (Hussin et al., 2018; Masson et al., 2025), and utility-scale new installations are expected to account for roughly two-thirds of this rapid expansion (Gianluca et al., 2025). Yet are rapidly growing solar farms inherently 'good'?

Having lived in Korea for 25 years and taken various courses on environment and energy, I was exposed to various technical knowledge about solar panels. However, I have never encountered explanations about possible side-effects associated with their installation. There remains a paucity of research on issues that do not readily appear in simple figures, and such issues are frequently overlooked. Therefore, in this essay, I would like to address the following question:

What needs to be considered in the planning and construction of solar farms?

To answer this, I will analyze solar farm projects in Honam region implemented during the '3020 Renewable Energy Implementation Project' (2017 to 2025). To this end, it poses three sub-questions: What happened in solar farms in Honam region between 2017 and 2025? From the perspectives of spatial justice, socio-technical, and (eco)systems, what has produced problems? How are these problems observed in other countries?

To address this question, I used Korean news sources and academic articles on policy, and, for the analysis through the three perspectives, I drew on relevant books and scholarly papers.

The remainder of this essay is organized as follows. Chapter 2 describes the project, Chapter 3 outlines the three analytical perspectives, and Chapter 4 analyzes the project's side effects through these perspectives. Chapter 5 reviews Dutch 'Zonnepark' and limitation of the essay, followed by Chapters 6, conclusion.

2. Renewable Energy 3020 Implementation Project

In 2017, Moon administration announced the ‘3020 Renewable Energy Implementation Project’ (hereafter ‘3020 Project’), setting a target of increasing the share of renewable energy from 7% to 20% by 2030 (Kim et al., 2021).

Studies have shown that the Honam region (comprising North and South Jeolla Province and Gwangju (Rowe et. al.,1993:1) (Figure 1)), with its abundant solar irradiation, is one of the most suitable areas for solar power (Choi & Kwak, 2008; Kim & Kim, 2018). Accordingly, solar power in Honam increased dramatically during ‘3020 Project’. In 2017, solar power generation in Honam amounted to 3,021.4 GWh. In 2023, it had risen to 13,255.8 GWh (Korea Energy Agency, 2024). This rapid surge in solar generation has made Honam a central hub for the deployment of solar energy. Moreover, in connection with RE100 commitments and the expansion of private investments, large corporate-led solar power complexes have been established, with both the scale and the pace of development increasing significantly (Ko, 2025; Kim, 2025).

However, this rapid and forced expansion of solar power under these conditions has generated considerable backlash. (Figure 2, 3, 4)

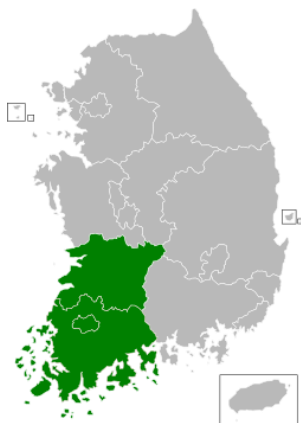


Figure 1 – Honam Region



Figure 2 – Sogum island solar farm (Kim, 2025)



Figure 3 – Solar panel at the mountain (Park, 2019)



Figure 4– Solar panel near the fields (Park, 2019)

3. Mechanism of Spatial Justice, Socio-technical and (Eco)system

Within the overall system we live in, people (e.g., policymakers, planners) introduce specific inputs about their interest (e.g. investment in technologies, '3020 project') to obtain outputs (e.g., increase in energy production from renewable sources). Influenced by the broader flows within the system, these inputs pass through a specific unknown function and emerge as outputs (Figure 5). However, for various reasons, intended outputs are not always achieved. They sometimes result in huge side-effects or strong opposition, and this is mainly due to three mechanisms that have been overlooked.

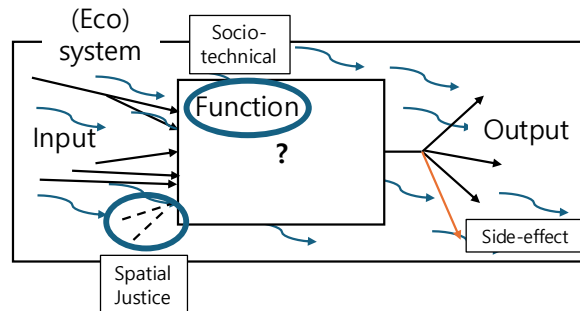


Figure 5 – Diagram of how 3 perspective works

First, the most straightforward reason is that not everyone is considered when inputs are introduced. In many cases, the voices of vulnerable groups or minorities are not reflected. This can be examined from the perspective of **Spatial Justice**. According to Rocco et. al. (2025), Spatial Justice can be classified into *distributive justice* (equitable distribution of opportunities), *procedural justice* (fairness of decision-making processes), and *recognition justice* (acknowledgement and respect for plurality). Moreover, each type of justice consists of three elements (Table 1). Through an analysis of Spatial Justice, it is possible to examine and confirm whether everyone's views have been reflected in the inputs and thereby identify what has been excluded. Excluded parts can incur two problems, displacement (when recognition claims crowd out redistribution, worsening inequality) and reification (when identities are treated as fixed and homogeneous, masking internal differences and power) (Fraser, 2000).

	Component	Description
Distributive Justice	Fair Allocation	Fair distribution of resources and benefits
	Improve Access to Opportunities	Improve affordable and available access to key opportunities
	Empower people to Adapt and Adopt	Enable people to actively adapt and adopt solutions
Procedural Justice	Democratic Engagement	Ensure ongoing public participation in decision-making
	Foster Internal Adaptiveness	Build flexible, learning-oriented internal processes
	Institutional Responsiveness	Respond fairly and transparently to external stakeholders
Recognition Justice	Legal Empowerment	Secure rights and dignity through legal frameworks
	Support for collective care practices	Support everyday collective care and solidarity practices
	Recognise and Foster the Pluriverse	Value and nurture diverse worlds, cultures and knowledges

Table 1 – 9 components of spatial justice (Rocco et. al., 2025)

Second, conflicts can also arise because the function is conceived in an overly simplified way by standardizing it or enforcing a resolute singularity, without accounting for the complexity of natural and social systems. This results from the *hubris of elites* and the *flaw* of their well-intentioned desire (Scott, 2020). Similar problems can be identified from Foucault's postmodern (technical) perspective. He argued that by *decreasing cost*, *increasing intensity*, and *increasing docility and utility*, disciplinary power can be rendered maximally efficient and pervasive (Foucault, 2012). Foucault shows how modern governance repeatedly seeks efficiency by making people and places legible and governable through standardization which made lots of failure at twentieth-century authoritarian countries

(Scott, 2020). In contrast to a technical perspective, **socio-technical** view can be defined as an effort to interpret 'function' through multiple lenses that incorporate social dimensions, thereby preventing technical missteps and generating outputs through a more socially inclusive conception. It allows us (while contingency within the function can never be fully eliminated) to evaluate the function more comprehensively and minimize side-effects of the output.

Lastly, **(eco)system's** broader flow strongly shapes every step from input to output. This system includes both the general characteristics of capitalism and institutional features of South Korea in the following case. Tsing (2012) introduces, against the flow, the concept of *non-scalability*, which creates fissures within systems that focus exclusively on scalability. In the current system planning tends to privilege elements that can be scaled, while non-scalable frictions are treated as noise and are externalized. Larsen (2008) likewise argues that sustainability must integrate social, economic, and environmental dimensions to occur simultaneously in each, yet contemporary policy and public discourse often treat sustainability as primarily environmental. (Figure 6, 7)

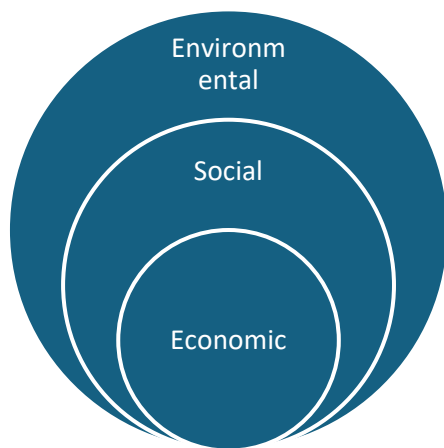


Figure 6 – Current perspective on sustainability

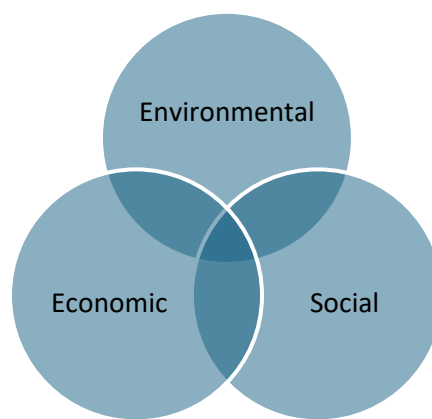


Figure 7 – Larsen's view of sustainability

4. Unexpected outputs of the 3020 Project – view from Spatial Justice, Socio-technical and (Eco)system.

Spatial Justice

The project input involved exclusions, in that certain actors, places, and concerns were left out of consideration. Resident briefings for solar farm installations were often held formalistic, so that residents, affected by the solar farm, were cut out of the loop. In an interview with *Nongmin Newspaper*, a farmer who had to relocate his fields due to a solar construction project, stated: “I only learned about the project after the construction had already begun. Who carefully checks the local newspaper or the city hall website bulletin board?” (Lee, 2025). In addition, laws and institutions were revised under the ‘3020 project’ in ways that made it more advantageous to expand solar power generation. In 2019, the government amended *Farmland Act* so that land previously used as farmland could be converted to solar farm if it was classified as saline-damaged land. At that time, salinity was measured at a depth of 30–60cm rather than at the 15cm, root depth relevant for rice cultivation. This led to numerous cases in which land, still being farmed normally, was classified as saline-damaged, which makes it easier to change to solar farms (Seo, 2021). [*Procedural Justice*]

Additionally, elderly farmers, who make up about 60% of the farming population, and tenant farmers, who cultivate roughly 50% of all farmlands (Lee et al., 2020), were easily excluded from access to information and opportunities for participation. SaGong and Jang (2020) point out that the spread of solar projects has led to forest degradation, ecological disturbance, and negative impacts on topography and landscape, and conflicts over the relocation of ancestral grave, yet institutional mechanisms to prevent and mediate these conflicts remain weak. Moreover, most of the electricity produced in these areas exceeded local demand and is transmitted to the Seoul Metropolitan Area. Therefore, residents of the installation areas bore the burdens of land-use while electricity was primarily consumed in distant areas (Yoo, 2025). [*Recognition Justice*]

Furthermore, in the project, economic benefits and environmental & social burdens are distributed asymmetrically. According to a survey of rural solar deployment, project revenues and land rents are accumulated primarily to developers and landowners, whereas peasants must bear various environmental externalities such as landscape degradation, crop damage related to electromagnetic waves and glare, and concerns about landslides. Additionally, maintenance systems and training on operation are insufficient, and documents cases where equipment installed experienced breakdowns or efficiency losses without systematic follow-up (Kim et al., 2018). [*Distributive Justice*]

Socio-technical

‘3020 Project’ clearly exemplifies a modernist technical view that tries to simplify the function of society and fails to incorporate social dimensions. The plan set quantitative targets and timetables and emphasized solar panel installations on a big scale for lowering the cost with high intensity by *resolute singularity, ignoring complexity and contingency* of the people and environment. Saemangeum mega-project, the largest solar farm project under the “3020 project,” likewise framed reclaimed coastal space as a single, standardized platform for gigawatt-scale solar build-out, privileging one-dimensional metric of capacity expansion while downplaying site-specific ecological constraints and long-standing local contestation around tidal-flat ecosystems and governance (Park, 2019). This simplification treated diverse rural lifeworld as interchangeable ‘sites’ for capacity expansion. Moreover, the government introduced *standardized* economic incentives, such as Feed-In Tariffs (cost-based compensation to renewable energy producers with long-term contracts) and

Renewable Portfolio Standards (quantity-based policy and requires energy producers to generate a certain level of renewable energy) (Park & Koo, 2018), to drive deployment, without considering the view of social and environmental effects. In this sense, the way policymakers construed the plan's function reflects a policy design that privileged national efficiency over local social and ecological concerns.

Similarly, during the project, *hubris of the planning elite* was evident in how the policy was conceived and executed. Central authorities assumed that installing the countryside with solar investment was an unequivocal good. It was a *well-intentioned desire* to install green energy, but one *with a fatal flaw*: the human factor. Little effort was made to meaningfully involve or even inform the residents during the plan's design. Yujin Lee, a researcher at the Green Transition Institute, said that "The lack of consideration about 'space and agents' during the design process of '3020 Project' has resulted in serious conflicts (on local communities and residents)." (Kang, 2022). This top-down rollout presumed simple technical 'function' without societal view that rural people would passively embrace the opportunity. Elites thought any resistance could be managed through compensation, underestimating farmers' attachment to their land.

(Eco)system

The side effects described above are deeply rooted in the structural features of Korean governance, particularly within Korea's dominant two-party system, where performance-oriented public-sector incentives emphasize rapid, quantifiable outputs, often privileging short-term deliverables over long-term system building (Im & Cha, 2024). Additionally, under Korea's five-year, single-term presidency, political incentives are front-loaded, and lame-duck dynamics can emerge early, intensifying pressure to demonstrate visible achievements within a single administration (OECD, 2019). Also, as administrations change, energy and climate initiatives are frequently reorganized or discontinued, weakening policy continuity and investment credibility across political cycles (Baydag, 2024; Yoon & Lee, 2017). In the flow of accelerated, target-driven implementation, land is rendered into standardized, isolated units that can be tightly controlled, with interchangeable modules that facilitate quick profit extraction, without accounting for ecosystem and local livelihoods. This dynamic is further reinforced within a growth-oriented scalable capitalist flow and within comparative evaluation regimes that rank through simplified numerical indicators on the paper.

Moreover, the '3020 Project', treated less as an integrated sustainability agenda across social, economic, and environmental dimensions and more as an environmental expansion program, can also be seen in its capacity-centered performance framing, where 'success' is narrated primarily through hitting quantitative renewable targets (Kim et. al., 2020). This performance framing exemplifies how Korea's broader growth-oriented, scalability-seeking system elevates easily countable megawatts as sustainability, while relegating the non-scalable social and economic frictions that is constitutive of sustainability to noise that is displaced or externalized.

These three perspectives are tightly coupled. Spatial injustice and socio-technical perspective emerge from bigger (eco)system flows. Under Korea's biparty system, projects face pressure to deliver fast, encouraging resolute singularity that amplifies 'well-intentioned desire with fatal flow' and ignores weaker voices. Within flow of capitalism, solar development is only focused on economical view through standardized permitting, contracts, and interchangeable land, enabling efficiency by excluding specific people and places from consideration, and in this way, forecloses even the possibility of reorienting toward degrowth, as articulated by Hickel (2020), which lies outside the dominant growth trajectory. Additionally, Spatial justice enriches and broadens the input of the

system, and socio-technical perspective opens the function so that the translation from inputs to outputs becomes capable of reducing side-effects. Once elites move beyond technical view and intentionally incorporate social dimensions, a wider range of people is automatically considered. In this sense, the expansion of the function and the expansion of inputs often occur simultaneously. For example, if policymakers avoid rigid standardization, listen to more diverse voices, and tailor solutions to each group instead of imposing one-size-fits-all measures, the number of people who are ignored will naturally decrease.

Therefore, to reduce side-effects that have arisen solar farms in Honam, drawing on an integrated analysis of these three perspectives is needed, slowing down and taking greater account of surrounding contexts.

5. Comparative Insights, Limitations, and Directions for Further Research

5.1. Beyond Honam, Zonnepark in the Netherlands

Side-effects of the solar farm are not unique to Honam. They also appear in the Netherlands, where solar provided about 16% of electricity in 2023, and around one fifth of capacity came from ground-mounted parks (Zonnepark) (Power Technology, 2024; VP Solar, 2024) (Figure 8 & Figure 9). As in Honam, conflicts emerge around land use, landscapes, and who bears costs versus benefits. However, various approaches have been explored to reduce such side-effects. Van der Zee et al. (2019) examine how solar parks can be designed to better integrate nature conservation and agricultural functions, and Becker et al. (2025) investigate spatial configurations in which solar installations are deliberately spaced to minimize negative impacts on farmers and surrounding landscapes. Nevertheless, research that simultaneously considers these three perspectives in an integrated manner remains limited - spatial justice to identify exclusions in inputs, socio-technical lens to reveal what the project 'function' simplifies, and (eco)system lens to diagnose the incentives driving speed and scale. Used together, they enable analysis of the root causes of side-effects and may even help steer the broader system's flow away from contemporary capitalism.

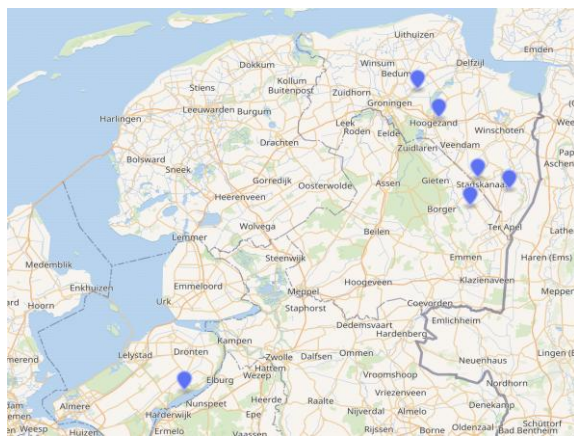


Figure 8 – location of Zonnepark in the Netherlands



Figure 9 – Zonnepark in the Netherlands (Mons, 2020)

5.2. Limitation & Further research recommendation

South Korea has begun to partially acknowledge problems and has undertaken efforts, such as creating formal venues for consultation with tenant farmers and introducing regional sunlight (community benefit-sharing) funds, to improve participation and local acceptance (Choi, 2025). Nevertheless, this essay foregrounds shortcomings to sharpen the diagnosis of the problem. In addition, given the limits of the research scope, it prioritizes an in-depth analysis of the current situation rather than proposing detailed remedies. Therefore, based on the problem analysis developed in this essay, further research is needed to comprehensively examine all aspects of shortcomings. This would enable the possibility to make more spatially just, socio-technically grounded, and capable of counteracting (eco)system flows.

6. Conclusion

This essay has shown that rapid solar expansion is not inherently good, and that a comprehensive analysis integrating three perspectives - spatial justice, socio-technical, and (eco)system - is needed. Building on such an analysis, solar parks should be developed in ways that minimize side-effects and take more people into account, slowing down and paying closer attention to surrounding contexts. '3020 Project' in Honam did not achieve this objective. The construction of solar parks was not informed by consideration of specific people, places, or environment. Rather, the focus was exclusively on maximizing speed and efficiency from a modernist technical perspective. This phenomenon is not exclusive to Korea. Rather, it is a global issue within the worldwide trend of expanding renewable energy.

7. References

- Baydag, R. M. (2024). Positions of Established and Emerging Powers Towards Climate Finance: The Cases of Germany and Korea. In *Emerging Trends in International Development and Climate Policy: A Focus on Korea and Germany* (pp. 39-52). Cham: Springer Nature Switzerland.
- Becker, F., Oudes, D., & Stremke, S. (2025). How does solar energy transform landscapes? A comparative spatial analysis of 46 built solar power plants in the Netherlands. *Renewable Energy*, 123621.
- Choi, B.-C., & Kwak, W.-S. (2008). A review of regional utilization rate analysis of photovoltaic power plants. *Proceedings of the Korean Society for New and Renewable Energy Conference*, 110–113. [in Korean]
- Choi, K.-H. (2025, June 10). “18 million won annually for a three-person household”: The tangible effects of solar and wind dividends. JoongAng Ilbo. <https://www.joongang.co.kr/article/25342782> [in Korean]
- Fraser, N. (2000). Rethinking recognition. *New left review*, 3, 107.
- Foucault, M. (2012). *Discipline and punish: The birth of the prison*. Vintage.
- Gianluca, T, Jad, B., Janne, L.P., Andrea, W., Jiayi, W., ... & Glem, W. (2025). *Renewables 2025-global status report*. Ren21, c/o UN Environment Programme, 158 ter rue du Temple, 75003 Paris (France).
- Hickel, J. (2020). *Less is more: How degrowth will save the world*. Random House.
- Hussin, F., Issabayeva, G., & Aroua, M. K. (2018). Solar photovoltaic applications: opportunities and challenges. *Reviews in Chemical Engineering*, 34(4), 503-528.
- Hussain, M. M., Concetti, C., Toke, D., Thomas, K., Duffy, P., & Vergunst, J. (2025). “Here comes the sun”: Determinants of solar farm planning at local authority level in England. *Energy Research & Social Science*, 120, 103916.
- Im, T., & Cha, S. (2024). The hierarchy-oriented bureaucracy of South Korea: a type of Neo-Weberian state?. *Journal of Policy Studies*, 39(2), 51-62.
- International Energy Agency. (2020). *Korea Renewable Energy 3020 Plan*. IEA Policies Database. <https://www.iea.org/policies/6569-korea-renewable-energy-3020-plan>
- Kang, Y. H. (2022, January 11). Why are farmers against solar panels in Korea? EnergyTransition.org. <https://energytransition.org/2022/01/why-are-farmers-against-solar-panels-in-korea/>
- Kim, C.-M. (2025, May 16). Solar power facility companies in the Honam region expand en masse into the Yeongnam region. Geoje Internet Newspaper. <https://www.gjn.kr/news/articleView.html?idxno=68469> [in Korean]
- Kim, J.-C., & Kim, K.-N. (2018). A study on region-specific optimal orientation and tilt angles of photovoltaic systems: A simulation case analysis using SAM. *New & Renewable Energy*, 14(2), 9–20. [in Korean]
- Kim, J.-H. (2025, November 19). Bright Energy Partners gains attention by expanding large-scale solar-based PPAs. HelloT Industrial News. <https://www.hellot.net/news/article.html?no=107339> [in Korean]
- Kim, Y.-J., Kim, S.-S., Chae, G.-S., Seo, D.-S., Park, J.-Y., Song, S.-H., & Chu, S.-M. (2018). A study on problems and improvement measures for the dissemination of rural solar power. *Korea Rural Economic Institute Policy Research Report*, 1–167. [in Korean]
- Kim, Y.-J., Seo, D.-S., Heo, J.-H., & Lee, J.-M. (2021). Carbon neutrality: Issues and challenges of rural solar power. *Korea Rural Economic Institute, Agricultural Policy Focus*, 1–21. [in Korean]
- Ko, G.-H. (2025, April 22). Jeonnam ranks highest nationwide in solar power generation and greenhouse gas reduction, emerging as a renewable energy hub. *Kyunghyang Shinmun*. <https://www.khan.co.kr/article/202504221203001> [in Korean]

- Korea Energy Agency, New & Renewable Energy Center. (2024.). Regional renewable energy generation by metropolitan region [Statistics]. Retrieved December 9, 2025, from https://www.knrec.or.kr/biz/statistics/supply/supply02_02_list.do
- Larsen, G. L. (2008). An inquiry into the theoretical basis of sustainability: Ten propositions. In *Understanding the social dimension of sustainability* (pp. 61-98). Routledge.
- Lee, M.-H., Kang, M., Kim, D.-H., Lee, J.-W., & Kang, D.-H. (2020). Measures to improve farmland ownership and use systems (Research report). Presidential Committee on Agriculture, Fisheries, and Rural Communities. <https://dams.pa.go.kr/dams/PUBLICATION/2025/10/21/DOC/SRC/0B04202510211010231410454019011.pdf> [in Korean]
- Lee, S.-N. (2025, February 3). Conflicts with residents intensify during solar installation—“communication urgently needed.” Nongmin Newspaper. <https://www.nongmin.com/article/20250203500902> [in Korean]
- Masson, G., Jäger-Waldau, A., Kaizuka, I., Lindahl, J., Donoso, J., & de L'Epine, M. (2025, June). A Snapshot of the Global PV Market and Industry. In *2025 IEEE 53rd Photovoltaic Specialists Conference (PVSC)* (pp. 0508-0510). IEEE.
- Organisation for Economic Co-operation and Development (OECD) Staff. (2019). *Understanding the drivers of trust in government institutions in Korea*. OECD.
- Park, I.-G. (2019, October 19). Civil society organizations in North Jeolla oppose the Saemangeum floating solar power project. Hankyoreh. <https://www.hani.co.kr/arti/area/honam/903316.html> [in Korean]
- Park, R., & Koo, M. G. (2018). South Korea's renewable energy policy. *Journal of International and Area Studies*, 25(2), 61-86.
- Park, S.-G. (2019, August 22). [View N] Mountains and fields covered with solar panels—environmentally destructive “green” energy. Hankook Ilbo. <https://www.hankookilbo.com/News/Read/201908211279061500> [in Korean]
- Power Technology. (2024, September 25). Solar PV in the Netherlands. Power Technology. <https://www.power-technology.com/data-insights/solar-pv-in-the-netherlands/>
- Rocco, R., Goncalves, J., Lopez, H., & Dąbrowski, M. (2025). The spatial justice benchmarking workshop guide (Version 3) [Workshop guide]. TU Delft Centre for the Just City.
- Rowe, S. M., Yoon, T. R., & Ryang, D. H. (1993). An epidemiological study of hip fracture in Honam, Korea. *International orthopaedics*, 17(3), 139-143.
- Sagong, J.-H., Jeong, O.-S., & Kwon, O.-S. (2018). Problems of photovoltaic power generation facilities and environmental response strategies. *Journal of the Korean Solar Energy Society*, 16(2), 13–20. [in Korean]
- Savini, F. (2023). Futures of the social metabolism: Degrowth, circular economy and the value of waste. *Futures*, 150, 103180.
- Scott, J. C. (2020). *Seeing like a state: How certain schemes to improve the human condition have failed*. Yale University Press.
- Seo, R. (2021, October 26). Prime farmland disguised as saline land—what will happen to farmers? Nongmin Newspaper. <https://www.nongmin.com/article/20211022346194> [in Korean]
- Tsing, A. L. (2012). On nonscalability: The living world is not amenable to precision-nested scales. *Common knowledge*, 18(3), 505-524.
- Yoon, J., Oh, Y., & Lee, J. D. (2017). The impact of policy consistency on technological competitiveness: A study on OECD countries. *Energy Policy*, 108, 425-434.
- Yoo, J.-H. (2025, July 15). Honam solar power increasingly transmitted to the Seoul metropolitan area. Maeil Business Newspaper. <https://www.mk.co.kr/news/business/11368646> [in Korean]

van der Zee, F., Bloem, J., Galama, P., Gollenbeek, L., van Os, J., Schotman, A., & de Vries, S.
(2019). Zonneparken natuur en landbouw (No. 2945). Wageningen, the Netherlands: Wageningen
Environmental Research.

VP Solar. (2024, January 13). Global market outlook 2023–2027: Netherlands.
<https://www.vpsolar.com/en/global-market-outlook-2023-2027-netherland/>

AI-Statement

During the writing process, ChatGPT 5.2 and ChatGPT 5.1 were used as a tool to check English spelling and grammar.